

Paperworks

1. CRF → CoNLL2003 Proposal Paper

- Features

1. **Current Word:** The token itself is the most fundamental feature, providing the base information for classification.
2. **Neighboring Words:** The words immediately before and after the current token are included as features. This context helps the model understand how the current token relates to its surroundings, which is especially important for disambiguating entities.
3. **Part-of-Speech (POS) Tags:** POS tags for the current word and neighboring tokens are included to provide grammatical context. This information helps the model distinguish between different entity types based on their syntactic roles.
4. **Chunk Tags:** Incorporating chunk information can further improve the model's ability to recognize multi-word entities by providing insights into phrase structure.
5. **Orthographic Features:** Features capturing the orthographic characteristics of words are also vital. These include:
 - o **Capitalization:** Whether the word is capitalized or in lowercase, as named entities often have specific capitalization patterns.
 - o **Digit Presence:** Indicates whether the token contains digits, which can be indicative of certain types of entities (e.g., dates or numerical values).
 - o **Length of the Word:** The number of characters in the word can provide additional information, as certain entities may have characteristic lengths.
6. **Word Shape:** This feature represents the pattern of capitalization, such as whether the word is in all caps, has an initial capital followed by lowercase letters, or is entirely in lowercase. This can help differentiate between entities like "NASA" and "nasa."
7. **Character N-grams:** Incorporating character-level n-grams can be useful for capturing morphological patterns, particularly for recognizing entities in noisy or informal text.

- Performance

Performance on CoNLL-2003

The CRF model's performance on the CoNLL-2003 dataset is represented in Table 1. This table summarizes the precision, recall, and F1-score for each named entity category, providing insights into the model's accuracy and robustness.

Entity Type	Precision	Recall	F1-Score
Person	92.3%	90.1%	91.2%
Location	91.7%	89.8%	90.7%
Organization	93.5%	92.4%	92.9%
Miscellaneous	85.0%	80.0%	82.4%
Overall	90.6%	88.2%	89.4%

Table 1: Performance of CRF Model on CoNLL-2003 Dataset

As seen in Table 1, the CRF model achieves an overall F1-score of 89.4%, with notable strengths in identifying Organizations (92.9% F1-score) and Persons (91.2% F1-score). This strong performance for specific entity types can be attributed to distinct syntactic and contextual clues that the CRF model can capture through its structured learning approach.

2. Survey paper (model analysis on CoNLL-2003)

- Features

2. Conditional Random Fields (CRF)

- Implemented as a linear-chain CRF, modeling the conditional probability of label sequences given the input sequence.
- Feature templates included:
 - ❖ Word-level: token identity, lowercased form, prefixes and suffixes (up to length 3).
 - ❖ Orthographic: capitalization, digit patterns, punctuation.
 - ❖ Contextual: neighboring words within a ± 2 window.
 - ❖ Linguistic: part-of-speech (POS) tags and chunk tags from a pre-trained parser.
- Training was performed using stochastic gradient descent with L2 regularization.

- Performance

Table 1. Performance comparison of NER algorithms on the CoNLL-2003 test set

Model	Precision	Recall	F1-score	Notes
Rule-Based	65.2	39.8	49.6	Strong precision, very low recall
CRF	87.8	84.1	85.9	Strong baseline, dependent on handcrafted features
BiLSTM-CRF	91.2	91.0	91.1	Robust contextual modeling with embeddings
BERT-base	92.5	92.3	92.4	Transformer-based, highest overall accuracy

3. Original CoNLL-2003 paper

McCallum ← CRF model

- Feature

	lex	pos	aff	pre	ort	gaz	chu	pat	cas	tri	bag	quo	doc
Florian	+	+	+	+	+	+	+	-	+	-	-	-	-
Chieu	+	+	+	+	+	+	-	-	-	+	-	+	+
Klein	+	+	+	+	-	-	-	-	-	-	-	-	-
Zhang	+	+	+	+	+	+	+	-	-	+	-	-	-
Carreras (a)	+	+	+	+	+	+	+	+	-	+	+	-	-
Curran	+	+	+	+	+	+	-	+	+	-	-	-	-
Mayfield	+	+	+	+	+	-	+	+	-	-	-	+	-
Carreras (b)	+	+	+	+	+	-	-	+	-	-	-	-	-
McCallum	+	-	-	-	+	+	-	+	-	-	-	-	-
Bender	+	+	-	+	+	+	+	-	-	-	-	-	-
Munro	+	+	+	-	-	-	+	-	+	+	+	-	-
Wu	+	+	+	+	+	+	-	-	-	-	-	-	-
Whitelaw	-	-	+	+	-	-	-	-	+	-	-	-	-
Hendrickx	+	+	+	+	+	+	+	-	-	-	-	-	-
De Meulder	+	+	+	-	+	+	+	-	+	-	-	-	-
Hammerton	+	+	-	-	-	+	+	-	-	-	-	-	-

- Performance

English test	Precision	Recall	$F_{\beta=1}$
Florian	88.99%	88.54%	88.76±0.7
Chieu	88.12%	88.51%	88.31±0.7
Klein	85.93%	86.21%	86.07±0.8
Zhang	86.13%	84.88%	85.50±0.9
Carreras (a)	84.05%	85.96%	85.00±0.8
Curran	84.29%	85.50%	84.89±0.9
Mayfield	84.45%	84.90%	84.67±1.0
Carreras (b)	85.81%	82.84%	84.30±0.9
McCallum	84.52%	83.55%	84.04±0.9
Bender	84.68%	83.18%	83.92±1.0
Munro	80.87%	84.21%	82.50±1.0
Wu	82.02%	81.39%	81.70±0.9
Whitelaw	81.60%	78.05%	79.78±1.0
Hendrickx	76.33%	80.17%	78.20±1.0
De Meulder	75.84%	78.13%	76.97±1.2
Hammerton	69.09%	53.26%	60.15±1.3
Baseline	71.91%	50.90%	59.61±1.2